MARKETING

Course-end Project 1

Restaurant Data Analysis, Exploratory Data Analysis & Recommendations

Project Write-up

This report documents the approach, data preparation, analysis and key findings from the restaurant consolidator project. The analysis was performed using Python, Pandas and visualization libraries, with the results intended to support an intelligent B2C restaurant recommendation portal.

Dataset: Restaurant dataset + Country-Code dataset

Records analysed: 9,551 restaurants | Final columns after country merge: 20

Prepared as a submission-ready project write-up

1. Project Overview

The objective of this project is to understand restaurant data before building an intelligent system that can identify star restaurants and generate useful recommendations for customers. The dataset contains information such as restaurant name, city, locality, cuisines, cost for two people, table booking, online delivery, price range, aggregate rating and votes. A separate Country-Code dataset was used to map country codes to country names.

I approached the project in stages rather than jumping directly into visualisation. First, I loaded and inspected the datasets. I then checked the structure, data types, missing values and duplicate records. After that, I explored geography, restaurant franchises, booking and delivery behaviour, cuisine patterns and rating-related variables. This helped me understand what the data can and cannot tell us about restaurant performance.

2. Tools and Technologies Used

- Python – primary language used for data analysis.
- Pandas – data loading, merging, cleaning, grouping and aggregation.
- NumPy – numerical operations.
- Matplotlib – bar-chart visualisation.
- Seaborn – imported for exploratory visualisation support.
- Tableau – planned/required dashboarding layer for interactive exploration.

The submitted source-code PDF shows the analysis workflow, including the data import, merge, inspection, missing-value checks, duplicate checks, grouping and cuisine analysis.

3. Data Import and Initial Inspection

The restaurant data was read from data.xlsx, while country information was read from Country-Code.xlsx. The country table contains 15 rows and 2 columns. A left merge was then performed on Country Code, producing the working DataFrame with 9,551 rows and 20 columns.

The original restaurant dataset had 19 attributes. After adding the country name from the country-code table, the working dataset contained 20 attributes. The data types included numeric fields such as longitude, latitude, average cost, price range, aggregate rating and votes, along with categorical fields such as city, cuisines, booking status, delivery status and rating text.

Item	Finding
Restaurant records	9,551
Attributes before country merge	19
Attributes after country merge	20
Country-Code records	15
Duplicate rows	0
Missing Restaurant Name	1
Missing Cuisines	9

4. Data Quality Assessment and Cleaning

The inspection identified two columns with missing values: Restaurant Name had 1 missing value and Cuisines had 9 missing values. All other listed fields had complete values in the merged dataset.

The duplicate check returned 0, meaning no exact duplicate rows were found in the working DataFrame. Therefore, there were no duplicate records to remove. This is important because the project asks for duplicate identification and removal; in this dataset, the correct action was to verify the absence of duplicates rather than artificially deleting records.

For cuisine analysis, the 9 missing cuisine values were temporarily replaced with the label Unknown. The cuisine field was then split on commas and exploded so that each individual cuisine could be counted separately. This transformation makes multi-cuisine restaurants usable for cuisine-level analysis.

5. Exploratory Data Analysis

5.1 Geographical Distribution

The city-level frequency analysis showed a strong concentration of restaurants in a small number of cities. New Delhi had the maximum number of restaurants with 5,473, followed by Gurgaon with 1,118 and Noida with 1,080. The minimum shown in the analysis was Miller, with only 1 restaurant.

The city bar chart in the submitted analysis also makes the concentration visible: New Delhi is much larger than the next group of cities. This geographical imbalance should be considered when using the dataset to build recommendations, because the model may have substantially more information about some locations than others.

5.2 Restaurant Franchising / National Presence

To examine franchise presence, I grouped restaurants by Restaurant Name and counted the number of distinct cities in which each name appeared. In the submitted result, Barbeque Nation appeared across 22 cities, followed by Pizza Hut across 12 cities and KFC across 8 cities. McDonald's, The Yellow Chilli, Pind Balluchi, Domino's Pizza and Mocha each appeared across 7 cities in the displayed output.

This measure is specifically a city-presence measure; it should not be interpreted as revenue, market share or profitability. It simply indicates how widely a restaurant name is represented across cities in this dataset.

5.3 Table Booking Analysis

The table-booking field contained 8,393 restaurants marked No and 1,158 marked Yes. The analysis calculated the ratio of restaurants allowing table booking to those not allowing it as 0.13797, approximately 0.138 : 1. In percentage terms, the 'Yes' group represents about 12.1% of the 9,551 restaurants.

5.4 Online Delivery Analysis

Online delivery was available for 2,451 restaurants, while 7,100 did not provide online delivery. The submitted calculation gives an online-delivery coverage of 25.66% of all restaurants in the dataset.

5.5 Votes: Delivery vs. Non-Delivery Restaurants

The total votes were grouped by online-delivery status. Restaurants without online delivery accumulated 980,731 votes, while restaurants with online delivery accumulated 517,914 votes. The absolute difference between the two totals was 462,817 votes.

This comparison describes the total number of votes in each group. It does not, by itself, prove that delivery availability causes higher or lower customer engagement because the groups have different numbers of restaurants and other characteristics may also influence voting.

6. Cuisine Analysis

6.1 Top Cuisine Entries

For cuisine analysis, restaurants with multiple cuisines were split into individual cuisine entries. The submitted bar chart shows the highest-frequency cuisine entries as North Indian, Chinese, Fast Food, North Indian, Chinese, Mughlai, Fast Food, Bakery, Cafe and Italian. The repeated labels indicate that the raw cuisine strings contain formatting variants that were not fully standardised before counting.

This is a useful data-cleaning observation: before using cuisine frequency as a production recommendation feature, values should ideally be stripped of leading/trailing spaces and standardised. The submitted analysis demonstrates the counting process, but the displayed result should be interpreted with this formatting limitation in mind.

6.2 Maximum and Minimum Cuisines per Restaurant

The cuisine string was split using commas and the number of cuisine entries per restaurant was calculated. The submitted output shows a maximum of 8 cuisines served by a restaurant and a minimum of 1.

A city-and-cuisine grouped table was also created using City and the exploded Cuisines field. For example, the displayed output includes cuisine counts for Abu Dhabi and Istanbul. This structure can be used to identify the most frequently represented cuisine within each city.

7. Cost and Rating Analysis

7.1 Distribution of Cost

The numerical summary shows that Average Cost for two has a median of 400 and a 75th percentile of 700 in the dataset's recorded currency units. The mean is approximately 1,199.21, while the maximum is 800,000. The large difference between the median and maximum indicates substantial variation and the presence of very high-cost observations.

Because the dataset contains multiple countries and currencies, cost values should not be directly compared across countries without currency conversion or appropriate country-level normalisation. This is an important consideration for any recommendation model.

7.2 Rating Distribution

The aggregate rating ranges from 0.0 to 4.9. The numerical summary reports a mean rating of 2.666, a median of 3.2, and a 75th percentile of 3.7. The dataset also contains Rating color and Rating text fields, which provide categorical interpretations of the numeric rating.

7.3 Factors That May Be Associated with Ratings

The available variables provide several reasonable dimensions for investigating rating differences. These include the number of cuisines offered, Average Cost for two, Price range, Has Table booking, Has Online delivery, Votes, City, Locality and Country. These variables can be examined together with Aggregate rating to identify patterns.

- Price / cost: customer expectations may differ across price levels, so rating patterns can be compared by price range.
- Online delivery: ratings can be compared between restaurants with and without delivery.
- Table booking: booking availability can be analysed against rating levels.
- Votes: vote volume provides a measure of customer engagement and can be compared with ratings.
- Cuisine variety: the number and type of cuisines can be used as restaurant-characteristic variables.
- Location: city, locality and country can capture geographic differences in restaurant markets.

The submitted notebook does not show a formal correlation matrix or statistical significance test for these relationships. Therefore, these variables should be treated as candidate explanatory factors rather than proven causes of ratings.

8. Dashboarding and Business Use

The final project requirement is to visualise the variables in Tableau so that a user can explore the restaurant landscape and understand characteristics associated with highly rated restaurants. A practical dashboard can combine KPI cards, filters and charts so the user can move from an overall market view to a specific city or restaurant.

Suggested dashboard structure

- KPI cards: total restaurants, average rating, total votes, online-delivery percentage.
- Geographical view: restaurants by country/city using location fields.
- City analysis: restaurant count by city with city and country filters.
- Cuisine view: top cuisines and cuisine variety by restaurant.
- Service features: table booking and online delivery comparison.
- Rating analysis: aggregate rating against price range, votes, cost and service options.

- Restaurant detail: a filtered table for restaurant name, city, cuisines, rating, votes, cost and services.

For a 'star restaurant' view, the dashboard should allow users to define their own threshold—for example, a high aggregate rating combined with meaningful vote volume—rather than relying on rating alone. This reduces the risk of treating a restaurant with very few votes as equivalent to one with substantial customer feedback.

9. Key Findings at a Glance

Metric	Result
Restaurants analysed	9,551
Cities represented	141
Maximum restaurant city	New Delhi – 5,473
Minimum restaurant city in output	Miller – 1
Most geographically widespread franchise in submitted result	Barbeque Nation – 22 cities
Table booking – Yes / No	1,158 / 8,393
Table-booking Yes : No ratio	0.13797
Online delivery – Yes / No	2,451 / 7,100
Restaurants with online delivery	25.66%
Votes – delivery Yes / No	517,914 / 980,731
Difference in total votes	462,817
Missing Restaurant Name	1
Missing Cuisines	9
Duplicate rows	0
Maximum cuisines per restaurant	8
Minimum cuisines per restaurant	1
Mean aggregate rating	2.666
Median aggregate rating	3.2
Maximum aggregate rating	4.9

10. Conclusion

This project provided a structured view of the restaurant dataset and demonstrated the complete basic EDA workflow: importing data, combining datasets, checking quality, identifying missing values and duplicates, transforming multi-valued cuisine data, grouping records and creating visual summaries.

The analysis shows a highly concentrated restaurant distribution by city, limited adoption of table booking and online delivery in the dataset, substantial variation in restaurant cost, and a wide range of cuisine combinations. It also highlights an important modelling issue: raw data quality and comparability matter. Cuisine formatting, multiple currencies and differences in the number of votes should be considered before building an automated recommendation system.

Overall, the analysis creates a useful foundation for the next stage of the project: an interactive Tableau dashboard and, ultimately, an intelligent recommendation framework that uses restaurant characteristics and customer feedback in a transparent and data-driven way.

Source of analysis: The findings in this write-up are based on the submitted project code/results and screenshots. No unsupported numerical results have been added where the submitted analysis did not provide them.

11. Submission Note

The accompanying screenshots document the Python analysis outputs, while the source-code PDF contains the corresponding notebook/code workflow. This write-up is intended to accompany those materials as the written explanation of the project.